

AEGIS — Setup Checklist

Personal environment setup tasks for building the MVP/POC. Complete these in order — each section unblocks the next.

GCP Production Status (2026-02-23): `aegis-prod-488120` is live. API: `https://api.aegisimaging.ai` — all services healthy, cloud smoke suite 11/11 PASS. DIMSE receiver: `aegis-prod-dimse-receiver` (Compute Engine VM, `us-central1-a`, static IP `35.232.172.221`, port 11112). CI/CD: Cloud Build triggers active in `us-central1` (`deploy-on-develop` + `terraform-apply-on-develop`).

Monitoring & Observability

Cloud Monitoring Dashboard (requires GCP console access):
<https://console.cloud.google.com/monitoring/dashboards?project=aegis-prod-488120>

- Alert policies** (9 active):
- API 5xx rate, API p99 latency, API uptime check
 - Cloud SQL CPU, disk, connections
 - Cloud Run memory
 - Study stuck SLA, pipeline failures

0. Business & Account Setup

Each LLC gets its own accounts. Do not share accounts across AEGIS Imaging LLC and Encore Music LLC — keeps billing, sender reputation, and access controls separate per legal entity.

Per-LLC accounts to create

Service	AEGIS Imaging LLC	Why separate
GCP	New project under AEGIS billing account	Billing tied to LLC, Terraform state isolated
Vercel	New account (free Hobby tier)	Separate deployment, billing, env vars
Brevo	New account (free: 300 emails/day)	Separate sender domain/reputation (@aegisimaging.ai VS @encoremusicapp.com)
Business bank account	Open for AEGIS Imaging LLC	Required for GCP billing, separates finances

GitHub — use Organizations (one personal account is fine)

- ☐ Create GitHub Organization `aegis-imaging` (free tier)
- ☒ Repo transferred to `aegis-imaging/aegis` — transfer complete

P.O. Box (shared across all LLCs)

- ☐ Get a P.O. Box (one box, shared by AEGIS Imaging LLC, Encore Music LLC, Matt Senjem Music LLC)
- ☐ Amend each LLC filing with MN Secretary of State (~\$35 each) to replace home address with P.O. Box

Prerequisites checklist

- ☐ AEGIS Imaging LLC EIN obtained ✓
- ☐ Open business bank account for AEGIS Imaging LLC
- ☒ Create GCP account + billing account for AEGIS Imaging LLC — project `aegis-prod-488120` , billing `016DEE-91CE5C-ECB970`
- ☐ Create Vercel account for AEGIS Imaging LLC
- ☐ Create Brevo account for AEGIS Imaging LLC (free tier: 300 emails/day)

1. Local Development Environment

- ☐ Install Go 1.24+ (`brew install go`)
- ☐ Install Terraform 1.5+ (`brew install terraform`)
- ☐ Verify Node.js 20+ and npm are installed (`node --version`)
- ☐ Install Docker Desktop (for building/testing containers locally)
- ☐ Clone the AEGIS repo and verify the monorepo structure

2. GCP Project Setup

- ☒ Create a new GCP project under the **AEGIS Imaging LLC billing account** — `aegis-prod-488120` (region `us-central1`)
- ☒ Link the AEGIS billing account to the project — `016DEE-91CE5C-ECB970`
- ☒ Install the gcloud CLI (`brew install google-cloud-sdk`)
- ☒ Authenticate: `gcloud auth login` and `gcloud auth application-default login`
- ☒ Set default project: `gcloud config set project aegis-prod-488120`

Beta launch auth (GCP IAP)

For the beta/MVP, use GCP Identity-Aware Proxy (IAP) to gate the admin dashboard. No custom auth flow needed — Google handles login.

User type	Auth	How
Upload portal (external sites)	None	Public — anyone with the link can upload
Admin dashboard (beta testers)	GCP IAP	Sign in with Google account

- ☒ Deploy Go API to Cloud Run — `https://api.aegisimaging.ai`
- ☒ Enable IAP on the Cloud Run load balancer
- ☒ **Provision the IAP service agent (one-time per project — run in Cloud Shell as project owner):**

```
gcloud beta services identity create \
  --service=iap.googleapis.com \
```

```
--project=YOUR_PROJECT_ID
```

This creates `service-{PROJECT_NUMBER}@gcp-sa-iap.iam.gserviceaccount.com`. Terraform grants it `roles/run.invoker` on the admin Cloud Run service automatically — but the identity must exist first. Safe to run multiple times (idempotent). Run this **before** `terraform apply` in section 4.

- ☒ Add beta testers' Google accounts to IAP access list: (`matthewsenjem@gmail.com`)

```
gcloud iap web add-iam-policy-binding \  
  --member="user:tester@gmail.com" \  
  --role="roles/iap.httpsResourceAccessUser"
```

- ☒ Add the same emails to `admin_users` table (role: `admin` or `viewer`) — `matthewsenjem@gmail.com` seeded via `FIRST_ADMIN_EMAIL`
- ☒ Set env vars on Cloud Run: `AUTH_ENABLED=true AUTH_PROVIDER=iap`
- ☒ Verify: tester visits admin dashboard URL → Google sign-in → dashboard loads

3. Terraform — Project Bootstrap

- ☒ Copy `terraform/project/terraform.tfvars.example` to `terraform/project/terraform.tfvars`
- ☒ Fill in your `project_id`, `region`, and `billing_account`
- ☒ Run `terraform init` in `terraform/project/`
- ☒ Run `terraform plan` and review the output
- ☒ Run `terraform apply` to enable all required GCP APIs
- ☒ Verify APIs are enabled: `gcloud services list --enabled`

4. Terraform — Infrastructure

- ☒ Copy `terraform/infra/terraform.tfvars.example` to `terraform/infra/terraform.tfvars`
- ☒ Fill in required `terraform/infra/terraform.tfvars` values:
 - `project_id = "aegis-prod-488120"`, `region = "us-central1"`, `environment = "prod"`
 - `api_domain = "api.aegisimaging.ai"`, `admin_domain = "admin.aegisimaging.ai"`
 - `iap_oauth_client_id`, `iap_oauth_client_secret`, `iap_access_members = ["user:matthewsenjem@gmail.com"]`
 - `db_password` (via Secret Manager), `db_password_secret_id`
 - image URLs for all services at `us-central1-docker.pkg.dev/aegis-prod-488120/aegis-services`
- ☒ Build and push images to Artifact Registry — all 8 services pushed at `:latest`
- ☒ Run `terraform init` in `terraform/infra/`
- ☒ Run `terraform fmt -check`
- ☒ Run `terraform validate`
- ☒ Run `terraform plan` and review
- ☒ Run `terraform apply` — created:

- VPC/subnet/private-service networking/Cloud NAT
- Artifact Registry repository (`aegis-services`)
- Cloud SQL PostgreSQL (private IP), Healthcare API dataset + DICOM stores, GCS buckets, Pub/Sub, BigQuery
- Cloud Run services (API + 6 sidecars + admin dashboard)
- Global HTTPS load balancer + managed cert + Cloud Armor + IAP admin backend

- ☒ Verify Artifact Registry repository exists:

```
gcloud artifacts repositories list --location=us-central1
```

- ☒ Verify Cloud Run services are deployed:

```
gcloud run services list --region=us-central1
```

- ☒ Verify staging bucket exists: `gsutil ls`
- ☒ Verify API health through LB domain:

```
curl -f https://api.aegisimaging.ai/healthz # returns {"status":"ok",...}
```

- ☒ Verify each sidecar health endpoint — all 6 sidecars (defacing, phi-detection, qc-service, bids-service, classification-service, protocol-service) return `{"status":"healthy"}` via `/health`
- ☒ Verify admin dashboard is gated by IAP — `https://admin.aegisimaging.ai` requires Google sign-in
- ☐ Verify SMTP egress static IP (not yet configured — email not enabled):

```
terraform output smtp_egress_ip
```

- ☐ Allowlist that IP on your SMTP relay/service
- ☐ Verify monitoring baseline exists:

```
gcloud monitoring policies list --format='value(displayName)'
```

- ☐ Open the Cloud Monitoring dashboard in GCP Console:

```
https://console.cloud.google.com/monitoring/dashboards?project=aegis-prod-488120
```

Expected: "AEGIS Operations — prod" dashboard with 11 tiles (API rate/errors/latency, Cloud Run instances/memory, Cloud SQL CPU/disk/connections, sidecar 5xx, pipeline failures, stuck-study SLA alerts).

- ☐ Confirm alert policies are active (9 total after Feature 55):

```
gcloud monitoring policies list --project=aegis-prod-488120 --format='table(displayName,enabled)'
```

- ☐ Confirm log-based metrics exist:

```
gcloud logging metrics list --project=aegis-prod-488120 --
format='value(name)'
# Expected: aegis-prod-pipeline-failures, aegis-prod-study-stuck
```

- ☐ See `terraform/monitoring/README.md` for full metric reference and runbook links.
- ☐ See `docs/runbooks/alert-response.md` for per-alert incident response procedures (triage steps, remediation commands, escalation paths for all 9 alert policies).

4x. DIMSE Receiver — Compute Engine VM

The DIMSE C-STORE SCP runs on a dedicated Compute Engine VM because Cloud Run cannot expose raw TCP ports. Terraform creates the VM; Cloud Build deploys new images by updating VM metadata and resetting the instance.

- ☒ VM `aegis-prod-dimse-receiver` created in `us-central1-a` via `terraform/infra/dimse.tf`
- ☒ Static regional IP `35.232.172.221` assigned; firewall rule allows TCP 11112 from `0.0.0.0/0`
- ☒ VPC-internal firewall allows TCP 8080 from Go API → DIMSE VM health endpoint
- ☒ Startup script mounts GCS staging bucket via `gcsfuse` at `/app/data` and starts the DIMSE container
- ☒ `dimse_receiver_image` and `dimse_api_url` set in `terraform/infra/terraform.tfvars` (stored in Secret Manager as `aegis-prod-terraform-tfvars` version 2)
- ☐ Verify DIMSE receiver is running after a test C-STORE from PACS:

```
# From any host with storescu installed
storescu -v -aec AEGIS 35.232.172.221 11112 /path/to/test.dcm
# Then verify in admin dashboard → Studies tab
```

- ☐ Verify VM health endpoint (VPC-internal only):

```
# From a Cloud Run service or Cloud Shell with VPC access
curl http://<vm-internal-ip>:8080/healthz
```

To update the DIMSE receiver image (done automatically by Cloud Build on `develop` push):

```
# Manual update (if needed):
gcloud compute instances add-metadata aegis-prod-dimse-receiver \
  --zone=us-central1-a \
  --metadata=dimse-image=us-central1-docker.pkg.dev/aegis-prod-488120/aegis-
services/dimse-receiver:latest
gcloud compute instances reset aegis-prod-dimse-receiver --zone=us-central1-a
```

4y. Cloud Build CI/CD Triggers

Cloud Build triggers were created by `scripts/gcp_setup_cloudbuild.sh` and run in `us-central1`.

- ☒ GitHub App connection established (Cloud Build → `aegis` connection → `aegis-imaging/aegis` repo)
- ☒ Trigger `deploy-on-develop` — fires on push to `develop` ; runs `cloudbuild.yaml` (builds+pushes all images, deploys Cloud Run services, hot-swaps DIMSE VM)
- ☒ Trigger `terraform-apply-on-develop` — fires when `terraform/infra/**` changes on `develop` ; runs `cloudbuild.terraform.yaml` ; reads `terraform.tfvars` from Secret Manager
- ☒ Cloud Build SA `aegis-cloud-build@aegis-prod-488120.iam.gserviceaccount.com` has all required IAM roles (tracked in `terraform/project/main.tf`)

Monitor recent builds:

```
gcloud builds list --project=aegis-prod-488120 --region=us-central1 --limit=5
```

Re-run setup (idempotent):

```
./scripts/gcp_setup_cloudbuild.sh
```

4a. First Admin Bootstrap

On the first deployment, the `admin_users` table is empty, so no one can log in to create admin users (chicken-and-egg). The API solves this with the `FIRST_ADMIN_EMAIL` env var — set it in `terraform/infra/terraform.tfvars` before `terraform apply` :

```
first_admin_email = "ops@aegisimaging.ai" # same as iap_access_members
```

Terraform sets `FIRST_ADMIN_EMAIL` on the API Cloud Run service. On startup, if `admin_users` is empty, the API seeds this email as the first admin (role: `admin` , enabled: `true`). Subsequent restarts are no-ops once any admin exists.

- ☒ Set `first_admin_email` in `terraform/infra/terraform.tfvars` before the first `terraform apply`
- ☒ Use the **same email address** as your first entry in `iap_access_members` so the user can log in immediately
- ☒ After apply, verify the admin was seeded by checking the API startup logs:

```
gcloud logging read 'resource.type="cloud_run_revision" AND
textPayload:"first-admin bootstrap: created admin user"' \
--project=YOUR_PROJECT_ID --limit=5 --format='value(textPayload)'
```

- ☒ Open the admin dashboard — `matthewsenjem@gmail.com` logs in, sees dashboard without "403 user not registered" error

To add more admins after the first login: use the **Users** tab in the admin dashboard, or call the API directly:

```
curl -X POST https://<api_domain>/api/admin-users \
-H "Authorization: Bearer <IAP_TOKEN>" \
-H "Content-Type: application/json" \
```

```
-d
'{"email":"colleague@example.com","name":"Alice","role":"admin","enabled":true}'
```

4b. Secrets Bootstrap and Rotation

GCP (Cloud SQL + Cloud Run API)

- ☒ Confirm DB password secret exists — `aegis-prod-db-password` in Secret Manager (`aegis-prod-488120`)
- ☒ Confirm API service account has secret accessor — verified via Terraform IAM binding
- ☐ Rotate DB password (dev drill — not yet done):

```
export NEW_DB_PASSWORD='<new-strong-password>'
gcloud secrets versions add aegis-dev-db-password --data-file==
<<<"$NEW_DB_PASSWORD"
gcloud sql users set-password aegis-api --instance=aegis-dev-postgres --
password="$NEW_DB_PASSWORD"
gcloud run services update aegis-api --region=us-central1 --update-env-
vars=ROTATION_EPOCH=$(date +%s)
```

- ☐ Verify post-rotation health:

```
curl -f https://<api_domain>/healthz
```

AWS (RDS managed master credentials)

- ☐ Confirm RDS is configured with managed master password in AWS Secrets Manager:

```
aws rds describe-db-instances --db-instance-identifier aegis-postgres \
--query 'DBInstances[0].MasterUserSecret.SecretArn' --output text
```

- ☐ Rotate AWS RDS master credentials (dev drill):

```
aws rds modify-db-instance --db-instance-identifier aegis-postgres \
--rotate-master-user-password --apply-immediately
```

4c. Automated Cloud Smoke Suite

- ☒ Run the cloud smoke suite against deployed API:

```
SSL_CERT_FILE=/etc/ssl/cert.pem python3 scripts/cloud_smoke_test.py \
--base-url https://api.aegisimaging.ai \
--iap-email matthewsenjem@gmail.com
```

- ☒ Verify suite exits with status code `0` and prints all PASS steps — **11/11 PASS in 2.56s** (2026-02-22):
 - `healthz` ✓ `auth.me` ✓ `admin.users.registered` ✓

- `upload.init` ✓ `upload.file` ✓ `upload.complete` ✓
 - `pipeline.progression` ✓ `study.approve` ✓
 - `share.create` ✓ `share.redeem` ✓ `share.download` ✓
- ☐ Verify fail-fast behavior:
 - re-run with an invalid admin header and confirm the suite fails quickly at `auth.me`
 - re-run with an invalid `--base-url` and confirm early transport failure

Manual trigger via GitHub Actions:

- Workflow: **Cloud Smoke** (`.github/workflows/cloud-smoke.yml`)
- Set input `base_url` and (optional) repository secret `CLOUD_SMOKE_ADMIN_HEADER`

4d. Terraform — AWS HTTPS + Cognito Edge/Auth

- ☐ Copy `terraform/aws/terraform.tfvars.example` to `terraform/aws/terraform.tfvars`
- ☐ Fill required values:
 - `aws_region` , `environment` , `project_name`
 - `acm_certificate_arn` (issued cert in the same region as ALB)
 - `cognito_domain_prefix` (region-unique)
 - optional callback/logout URL overrides
- ☐ Run:

```
terraform -chdir=terraform/aws init
terraform -chdir=terraform/aws fmt -check
terraform -chdir=terraform/aws validate
terraform -chdir=terraform/aws plan
terraform -chdir=terraform/aws apply
```

- ☐ Verify HTTP to HTTPS redirect:

```
curl -I http://<alb_dns>
```

- Expect `301` redirect to `https://...`
- ☐ Verify unauthenticated protected path triggers Cognito auth:
 - Open `https://<alb_dns>/api/studies` in an incognito browser
 - Expect redirect/challenge to Cognito hosted UI
- ☐ Verify public path bypass remains available (for system health):

```
curl -f https://<alb_dns>/healthz
```

4e. Terraform — Azure Infrastructure

Prerequisites

- ☐ Install Azure CLI: `brew install azure-cli` (macOS)
- ☐ Log in: `az login`

- ☐ Set subscription: `az account set --subscription <SUBSCRIPTION_ID>`
- ☐ Install Terraform: `brew install terraform`

One-time: Create Terraform state storage

```
az group create --name aegis-tfstate --location eastus
az storage account create \
  --name aegistfstate \
  --resource-group aegis-tfstate \
  --sku Standard_LRS \
  --allow-blob-public-access false
az storage container create \
  --name tfstate \
  --account-name aegistfstate
```

One-time: Create OIDC federated service principal for GitHub Actions

```
# Create service principal
SP=$(az ad sp create-for-rbac --name aegis-github-actions \
  --role Contributor \
  --scopes /subscriptions/<SUBSCRIPTION_ID> \
  --output json)

echo "Client ID:      $(echo $SP | jq -r .appId)"
echo "Tenant ID:      $(az account show --query tenantId -o tsv)"
echo "Subscription:    <SUBSCRIPTION_ID>"

# Add federated credentials for the develop branch
APP_ID=$(echo $SP | jq -r .appId)
az ad app federated-credential create \
  --id $APP_ID \
  --parameters '{"name":"aegis-develop","issuer":"https://token.actions.githubusercontent.com","subject":"repo:aegis-imaging/aegis:ref:refs/heads/develop","audiences":["api://AzureADTokenExchange"]}'
```

Add the following as **GitHub Secrets** on the repository:

- `AZURE_CLIENT_ID` — service principal App ID
- `AZURE_TENANT_ID` — Azure AD tenant ID
- `AZURE_SUBSCRIPTION_ID` — subscription ID
- `AZURE_ACR_REGISTRY` — set after Terraform apply (see outputs)
- `AZURE_RESOURCE_GROUP` — e.g. `aegis-prod`

Add as **GitHub Variables**:

- `AZURE_WEASIS_URL` — Weasis container app URL (set after first deploy)
- `AZURE_API_URL` — API container app URL (set after first deploy)

Apply Terraform

- ☐ Copy `terraform/azure/terraform.tfvars.example` to `terraform/azure/terraform.tfvars`
- ☐ Fill required values:
 - `azure_ad_tenant_id` (`az account show --query tenantId -o tsv`)
 - `db_admin_password` (strong password, ≥ 16 chars)
 - `alert_email`
 - `first_admin_email`
 - `api_domain` and `admin_domain` (optional — leave empty to use default ACA hostnames)
- ☐ Run:

```
terraform -chdir=terraform/azure init
terraform -chdir=terraform/azure fmt -check
terraform -chdir=terraform/azure validate
terraform -chdir=terraform/azure plan
terraform -chdir=terraform/azure apply
```

- ☐ Note the outputs:

```
terraform -chdir=terraform/azure output
```

Key outputs: `acr_login_server` , `api_url` , `admin_dashboard_url` , `acs_smtp_host`

First image push

```
# Build and push images manually for first deploy (before GitHub Actions is
configured)
ACR=$(terraform -chdir=terraform/azure output -raw acr_login_server)
az acr login --name $ACR

# Build all images
docker build --platform linux/amd64 -t $ACR/api:latest api/
docker build --platform linux/amd64 \
  --build-arg VITE_WEASIS_BASE_URL=<weasis_url> \
  --build-arg VITE_API_BASE_URL=<api_url> \
  -t $ACR/admin-dashboard:latest frontend/admin-dashboard/
# ... repeat for all 13 services

docker push $ACR/api:latest
# ... push all
```

Email (Azure Communication Services)

- ☐ After `terraform apply` , note the ACS SMTP config from outputs:
 - Host: `smtp.azurecomm.net` , Port: `587`
 - Username format: `<EntraAppClientId>|<TenantId>|<AcsResourceName>`
 - Password: OAuth2 access token (short-lived; use the ACS connection string for simpler SMTP)

- ☐ Alternatively: get the ACS connection string from Azure Portal → Communication Services → Keys
 - Use connection string auth for simpler SMTP integration with standard relay tools
- ☐ Set SMTP env vars on the API Container App:

```
az containerapp update --name aegis-prod-api \
  --resource-group aegis-prod \
  --set-env-vars \
    SMTP_HOST=smtp.azurecomm.net \
    SMTP_PORT=587 \
    SMTP_FROM=noreply@aegisimaging.ai \
    SMTP_USERNAME="<<EntraAppClientId>|<TenantId>|<AcsResourceName>" \
    SMTP_PASSWORD="<<oauth-token-or-access-key>"
```

Verify deployment

- ☐ API health check:

```
curl -f https://<api_fqdn>/healthz | jq
```

- ☐ Auth test (Easy Auth injects header automatically when accessing admin dashboard):
 - Open admin dashboard URL in browser → Azure AD login → redirects back
 - Your Azure AD email must be added to `admin_users` table (via `FIRST_ADMIN_EMAIL` var)
- ☐ Cross-cloud routing test: add Azure→GCP destination in admin dashboard, upload a study, verify STOW-RS forward

5. Sample DICOM Data for Local Testing

- ☐ Download sample brain MRI DICOM files for testing (options below):
 - TCIA (The Cancer Imaging Archive): <https://www.cancerimagingarchive.net/>
 - dicom.offis.de sample files: <https://dicom.offis.de/dcmstk/>
 - NEMA WG-6 sample DICOM: <ftp://medical.nema.org/medical/dicom/DataSets/>
- ☐ Place sample files in a local test directory (not committed to git)
- ☐ Verify files open in a DICOM viewer (e.g., Horos, 3D Slicer) to confirm validity

6. Frontend — Upload Portal

- ☐ `cd frontend/upload-portal && npm install`
- ☐ `npm run dev` — verify it runs on <http://localhost:3000>
- ☐ Install DICOM libraries: `npm install dcmjs dicom-parser`
- ☐ Test parsing a sample DICOM file in the browser console

7. Frontend — Admin Dashboard

- ☐ `cd frontend/admin-dashboard && npm install`
- ☐ `npm run dev` — verify it runs on <http://localhost:3001>
- ☐ Click **View** on any study row → Weasis viewer iframe appears inline

- ☐ Click **Open in new tab** ↗ → viewer opens in a new browser tab
- ☐ Click **Audit Log** tab → shows event table (empty until actions are taken)
- ☐ For a defaced head study: click **Review defacing** → side-by-side Weasis panel (Before/After)
- ☐ Click **Routing** tab → Destinations and Rules sections load (empty state)
- ☐ Click **Institutions** tab → Institutions table loads (empty state)

7a. Routing Rules Engine

- ☐ Open admin dashboard → **Routing** tab
- ☐ Create a Destination (type: `dicomweb` , any URL)
- ☐ Create a Routing Rule (e.g. `modality=MRI` → `auto_approve`) → appears in priority-ordered table
- ☐ Upload an MRI study via Upload Portal → study status auto-set to `approved`
- ☐ Verify routing log: `GET http://localhost:8080/api/studies/{id}/routing-log`
- ☐ Disable/enable rule toggle works; delete cleans up

7b. Institution Management

- ☐ Open admin dashboard → **Institutions** tab
- ☐ Create an institution (type: `sender`)
- ☐ Click **Projects** button on the row → inline project-link panel expands
- ☐ Link institution to a project: enter project UUID from `GET /api/projects` , choose role
- ☐ Verify link appears in table; unlink removes it
- ☐ Edit institution details; disable/enable toggle works

7c. Anonymization Profiles

- ☐ Open admin dashboard → **Profiles** tab
- ☐ Create a profile: name `Research` , project `default` , retained tags `PatientAge` , `StudyDate` → appears in table
- ☐ Click **Set default** → badge "default" appears on the row
- ☐ Upload a DICOM via the Upload Portal → check the de-identified tag diff: `PatientAge` and `StudyDate` should show action `K` (kept)
- ☐ Clear the default; re-upload → those tags are stripped again (normal Basic Profile)
- ☐ Edit / delete the profile

7d. Email Digest Subscriptions

- ☐ Start API with Mailpit enabled (`SMTP_HOST=localhost SMTP_PORT=1025 go run .`)
- ☐ Open admin dashboard → **Notifications** tab
- ☐ Create a subscription: email `test@example.com` , project `default` , frequency `weekly`
- ☐ In the DB, force a digest due: `UPDATE digest_subscriptions SET last_sent_at = now() - interval '8 days'`
- ☐ Restart the API → scheduler fires on startup → check Mailpit at <http://localhost:8025> for the digest email
- ☐ Verify subject: `AEGIS Weekly Summary – default – <date range>` , body has study counts, no PHI/UIDs

- ☐ Delete the subscription from the Notifications tab

7e. Admin Users

- ☐ Open admin dashboard → **Users** tab
- ☐ Create a user: email `admin@example.com` , name `Test Admin` , role `admin` → appears in table
- ☐ Edit → change role to `viewer` , add notes → save
- ☐ Disable the user → row greys out; re-enable
- ☐ Delete with confirm prompt
- ☐ Verify `admin_user.created` , `admin_user.updated` , `admin_user.deleted` appear in Audit Log tab

7f. Project Settings

- ☐ Open admin dashboard → **Projects** tab
- ☐ Click **+ New project** → enter name only (slug auto-generated) → create → appears in table
- ☐ Click **Edit** on the row → change description → save → description updates
- ☐ Try editing slug → warning hint is shown
- ☐ Verify `project.created` and `project.updated` in Audit Log tab
- ☐ GET `http://localhost:8080/api/projects/{id}` returns the updated project

7g. Upload Portal Features

Project selector

- ☐ Create a second project in admin dashboard → Projects tab → **+ New project** (e.g. name: "Research")
- ☐ Open upload portal at <http://localhost:3000>
- ☐ Verify project dropdown appears with both projects listed
- ☐ Select the new project → upload a study → verify study appears under that project in admin dashboard
- ☐ Delete the second project → reload upload portal → dropdown is hidden (single project auto-selected)

Drag-and-drop polish

- ☐ Drag a folder of DICOMs onto the drop zone → parsing starts with file count and total size displayed
- ☐ Click **Cancel** during parsing → returns to file selection
- ☐ Re-select files → on the preview screen, verify file count and size shown in the blue info bar
- ☐ Verify the **Confirm** button shows file count and size (e.g. "Confirm anonymization & upload (42 files, 156 MB)")
- ☐ During upload, verify current filename appears below the progress bar
- ☐ Click **Cancel** during upload → returns to preview with error message "Upload cancelled."

Email validation

- ☐ On the preview screen, type an invalid email (e.g. "foo") → tab away → red validation error appears

- ☐ Verify **Confirm** button is disabled while email is invalid
- ☐ Clear the email field → button re-enables (email is optional)
- ☐ Enter a valid email → validation error disappears, helper text shows "You'll receive an email when your study is approved or rejected"

Multi-study upload

- ☐ Prepare a folder containing DICOM files from 2+ different studies (different StudyInstanceUIDs)
- ☐ Upload the folder → portal detects multiple studies and shows a yellow notice
- ☐ Each study has its own StudySummary card with series/image count
- ☐ Click **Confirm & upload N studies** → each study uploads as a separate session
- ☐ Upload progress shows "Study X of N" during multi-study upload
- ☐ Success screen lists all session IDs and study UUIDs
- ☐ Verify each study appears as a separate row in the admin dashboard

Auto-retry

- ☐ (Optional) Throttle network in DevTools mid-upload → verify retries up to 3x before error

7h. Burned-in PHI Detection

- ☐ Install Tesseract OCR: `brew install tesseract` (macOS) or `apt-get install tesseract-ocr` (Linux)
- ☐ Start the PHI detection service:

```
cd phi-detection && pip install -r requirements.txt
uvicorn app.main:app --port 8082
```

- ☐ Verify health: `curl http://localhost:8082/healthz` → should show `{"status":"ok","backend":"tesseract"}`
- ☐ Start the Go API with PHI detection enabled:

```
cd api && PHI_DETECTION_SERVICE_URL=http://localhost:8082 go run .
```

- ☐ Open admin dashboard → **Routing** tab → create a rule: action `require_phi_scan` (any modality)
- ☐ Upload a study via the Upload Portal → study row shows PHI Scan badge: **pending**
- ☐ Click **Scan for PHI** → badge changes to **scanning** → then **clean** or **flagged**
- ☐ Check Audit Log tab → `phi_scan.triggered` and `phi_scan.complete` entries appear
- ☐ Verify admin can still Approve a flagged study (flag is informational, not blocking)

7i. Batch Import CLI

- ☐ Build the CLI: `cd api && go build -o aegis-import ./cmd/import`
- ☐ Dry run (absolute path required): `./aegis-import --dir /absolute/path/to/test-data/brain-mri --project default --dry-run`
- ☐ Verify dry run output shows study count, modality, body part for each study

- ☐ Full import: `./aegis-import --dir /absolute/path/to/test-data/brain-mri --project default`
- ☐ Verify studies appear in admin dashboard with `source=internal`
- ☐ Verify `import.batch` entries in Audit Log tab
- ☐ Re-run the same import → should report duplicate errors (StudyInstanceUID unique constraint)
- ☐ External provenance test: `source=external` requires canonical institution selector (`institution_id` or `institution_slug`)
- ☐ Deprecated field test: posting unknown fields (for example `institution_ae_title`) to `/api/import/batch` returns HTTP 400
- ☐ Test API endpoint:

```
curl -X POST http://localhost:8080/api/import/batch \
  -H "Content-Type: application/json" \
  -d '{"dir":"/absolute/path/to/test-data/brain-mri","project_slug":"default","dry_run":true}'
```

7j. QC Automation Service

- ☐ Start the QC service:

```
cd qc-service && pip install -r requirements.txt
uvicorn app.main:app --port 8083
```

- ☐ Verify health: `curl http://localhost:8083/healthz` → should show `{"status":"ok","backend":"basic"}`
- ☐ Start the Go API with QC enabled:

```
cd api && QC_SERVICE_URL=http://localhost:8083 go run .
```

- ☐ Open admin dashboard → **Routing** tab → create a rule: action `require_qc_check` (any modality)
- ☐ Upload a study via the Upload Portal → study row shows QC badge: **pending**
- ☐ Click **Run QC** → badge changes to **checking** → then **pass**, **warn**, or **fail**
- ☐ Check Audit Log tab → `qc_check.triggered` and `qc_check.complete` entries appear
- ☐ Verify admin can still Approve a study with warn/fail QC status (status is informational)

7k. NIfTI/BIDS Conversion Service

- ☐ Install dcm2niix: `brew install dcm2niix` (macOS) or `apt-get install dcm2niix` (Linux)
- ☐ Start the BIDS service:

```
cd bids-service && pip install -r requirements.txt
uvicorn app.main:app --port 8084
```

- ☐ Verify health: `curl http://localhost:8084/healthz` → should show `{"status":"ok","backend":"dcm2niix"}`
- ☐ Start the Go API with BIDS service enabled:

```
cd api && BIDS_SERVICE_URL=http://localhost:8084 go run .
```

- ☐ Open admin dashboard → **Routing** tab → create a rule: action `require_bids_conversion` (any modality)
- ☐ Upload a study via the Upload Portal → study row shows BIDS badge: **pending**
- ☐ Click **Convert to BIDS** → badge changes to **converting** → then **complete**
- ☐ Click **Download BIDS** → browser downloads a zip archive
- ☐ Unzip and verify BIDS structure: `dataset_description.json` , `participants.tsv` , `sub-*/anat/*.nii.gz` + `/*.json`
- ☐ Check Audit Log tab → `bids_conversion.triggered` and `bids_conversion.complete` entries appear

7l. Metadata Classification Service

- ☐ Start the classification service:

```
cd classification-service && pip install -r requirements.txt
uvicorn app.main:app --port 8085
```

- ☐ Verify health: `curl http://localhost:8085/healthz` → should show `{"status":"ok","backend":"heuristic"}`
- ☐ Start the Go API with classification service enabled:

```
cd api && CLASSIFICATION_SERVICE_URL=http://localhost:8085 go run .
```

- ☐ Open admin dashboard → **Routing** tab → create a rule: action `require_classification` (any modality)
- ☐ Upload a study with missing modality → study row shows Classification badge: **pending**
- ☐ Click **Classify** → badge changes to **classifying** → then **classified**
- ☐ Verify study modality and body_part updated from "-" to classified values
- ☐ Verify routing rules re-evaluated (e.g. a `require_defacing` rule for HEAD now fires)
- ☐ Check Audit Log tab → `classification.triggered` and `classification.complete` entries appear

7m. MRI Protocol Compliance Service

- ☐ Start the protocol service:

```
cd protocol-service && pip install -r requirements.txt
uvicorn app.main:app --port 8086
```

- ☐ Verify health: `curl http://localhost:8086/healthz` → should show `{"status":"ok","backend":"basic"}`

- ☐ Start the Go API with protocol service enabled:

```
cd api && PROTOCOL_SERVICE_URL=http://localhost:8086 go run .
```

- ☐ Open admin dashboard → **Protocol Templates** tab
- ☐ Create a template: name `ADNI4 T1w` , project `default` , manufacturer `SIEMENS` , sequence type `T1w_MPRAGE` , add rules (e.g. RepetitionTime target 2300, tolerance 5%, severity warning)
- ☐ Open admin dashboard → **Routing** tab → create a rule: action `require_protocol_check` (any modality)
- ☐ Upload a study via the Upload Portal → study row shows Protocol badge: **pending**
- ☐ Click **Check Protocol** → badge changes to **checking** → then **compliant**, **minor_deviations**, or **non_compliant**
- ☐ Check Audit Log tab → `protocol_check.triggered` and `protocol_check.complete` entries appear with per-parameter findings
- ☐ Verify admin can still Approve a non-compliant study (status is informational)
- ☐ Edit/delete the template from the Protocol Templates tab

7n. Authentication Middleware

Auth is disabled by default (`AUTH_ENABLED=false`) — all admin endpoints auto-authenticate as `ai@aegisimaging.ai` .

Dev mode (default)

- ☐ Start the API: `cd api && go run .`
- ☐ Verify auth identity: `curl -s http://localhost:8080/api/auth/me | jq .` → shows `ai@aegisimaging.ai` , role `admin`
- ☐ Verify public routes work without auth: `curl -s http://localhost:8080/healthz` → `ok`
- ☐ Verify admin routes work without auth headers: `curl -s http://localhost:8080/api/studies | jq .total`
- ☐ Check Audit Log tab → audit entries show `ai@aegisimaging.ai` as the actor (not `admin`)

Production mode (GCP IAP)

- ☐ Start API with auth enabled:

```
cd api && AUTH_ENABLED=true AUTH_PROVIDER=iap go run .
```

- ☐ Verify unauthenticated request is blocked:

```
curl -s http://localhost:8080/api/studies | jq .
# → {"error":"missing GCP IAP authentication header"}
```

- ☐ Verify unknown user is rejected:

```
curl -s -H "X-Goog-Authenticated-User-Email:
accounts.google.com:unknown@test.com" \
```

```
http://localhost:8080/api/studies | jq .  
# → {"error":"user not registered: unknown@test.com"}
```

- ☐ Register a user and verify access:

```
# First disable auth temporarily to create user  
cd api && go run .  
curl -s -X POST http://localhost:8080/api/admin-users \  
  -H "Content-Type: application/json" \  
  -d '{"email":"admin@test.com","name":"Test Admin","role":"admin"}'  
# Restart with auth enabled  
AUTH_ENABLED=true AUTH_PROVIDER=iap go run .  
curl -s -H "X-Goog-Authenticated-User-Email:  
accounts.google.com:admin@test.com" \  
  http://localhost:8080/api/studies | jq .total  
# → 0 (or study count)
```

Production mode (Azure AD)

- ☐ Start API with Azure AD auth:

```
cd api && AUTH_ENABLED=true AUTH_PROVIDER=azure go run .
```

- ☐ Verify Azure header auth works:

```
curl -s -H "X-MS-CLIENT-PRINCIPAL-NAME: admin@test.com" \  
  http://localhost:8080/api/studies | jq .total
```

Production mode (AWS ALB + Cognito)

- ☐ Start API with AWS ALB auth:

```
cd api && AUTH_ENABLED=true AUTH_PROVIDER=aws go run .
```

- ☐ Verify AWS ALB header auth works (simulate ALB-injected JWT):

```
# Create a base64url-encoded JWT payload with email claim  
PAYLOAD=$(echo -n '{"email":"admin@test.com"}' | base64 | tr '+/' '-_' | tr -  
d '=')  
JWT="header.${PAYLOAD}.signature"  
curl -s -H "X-Amzn-Okta-Data: ${JWT}" \  
  http://localhost:8080/api/studies | jq .total
```

AWS S3 Storage

- ☐ Start API with S3 storage (LocalStack for dev):

```
# Start LocalStack (S3-compatible)
docker run -p 4566:4566 localstack/localstack
# Create bucket
aws --endpoint-url=http://localhost:4566 s3 mb s3://aegis-dev
# Run API with S3 backend
cd api && STORAGE_MODE=s3 S3_BUCKET=aegis-dev S3_REGION=us-east-1
S3_ENDPOINT=http://localhost:4566 go run .
```

- ☐ Upload a study via the Upload Portal → verify files stored in S3 bucket
- ☐ View study in Weasis → verify DICOMweb proxy retrieves from S3
- ☐ Verify signed URLs work: upload + download flows complete without error

Azure AD App Registration Setup (for production Azure deployments)

1. Register the application in Azure Portal:

- Go to Azure Portal → Microsoft Entra ID → App registrations → New registration
- Name: AEGIS Admin Dashboard
- Supported account types: "Accounts in this organizational directory only" (Single tenant)
- Redirect URI: `https://<your-app-url>/.auth/login/aad/callback`
- Click Register

2. Configure authentication:

- Go to the app registration → Authentication
- Add platform: Web
- Redirect URIs: `https://<your-app-url>/.auth/login/aad/callback`
- Check "ID tokens" under Implicit grant

3. API permissions:

- Microsoft Graph → User.Read (delegated) — to read user email
- Grant admin consent for the directory

4. Configure the hosting platform:

- **Azure App Service:** Go to Settings → Authentication → Add identity provider → Microsoft → Select the app registration
- **Azure Application Gateway + AAD:** Configure AAD authentication on the gateway; it injects `X-MS-CLIENT-PRINCIPAL-NAME`
- **Self-hosted with MSAL.js:** Add MSAL.js to the frontend; backend validates JWT Bearer tokens (future enhancement)

5. Add authorized users:

- In the AEGIS admin dashboard Users tab, add each Azure AD user's email as an admin user
- Their Azure AD login email must match the `admin_users.email` field (case-insensitive)

6. Environment variables:

```
AUTH_ENABLED=true
AUTH_PROVIDER=azure # or "auto" to support both IAP and Azure
```

7o. RBAC Enforcement (Viewer Role)

The viewer role is read-only — viewers can browse all data but cannot create, update, delete, or trigger processing.

Backend verification

- ☐ Create a viewer user (with API running in default dev mode):

```
curl -s -X POST http://localhost:8080/api/admin-users \
  -H "Content-Type: application/json" \
  -d '{"email":"viewer@aegisimaging.ai","name":"Test Viewer","role":"viewer","enabled":true}'
```

- ☐ Restart API as viewer: `cd api && DEV_USER_EMAIL=viewer@aegisimaging.ai go run .`
- ☐ Verify read endpoints work:

```
curl -s http://localhost:8080/api/studies | jq .total    # → 200 OK
curl -s http://localhost:8080/api/institutions | jq .    # → 200 OK
curl -s http://localhost:8080/api/auth/me | jq .role    # → "viewer"
```

- ☐ Verify write endpoints are blocked:

```
curl -s -X POST http://localhost:8080/api/projects \
  -H "Content-Type: application/json" \
  -d '{"name":"test"}' | jq .
# → {"error":"insufficient permissions: requires admin role"}
```

Frontend verification

- ☐ Open admin dashboard at <http://localhost:3001> as viewer
- ☐ Verify Users tab is NOT visible in navigation
- ☐ Verify Studies tab: View + Download BIDS visible; Approve/Reject/Share/processing buttons hidden
- ☐ Verify Routing tab: data visible; Add/Edit/Delete/Toggle buttons hidden
- ☐ Verify Institutions tab: data visible; Add/Edit/Delete/Toggle/Link/Unlink hidden; Projects view button visible
- ☐ Verify Profiles, Protocol Templates, Notifications, Projects tabs: data visible; create/edit/delete buttons hidden
- ☐ Switch back to admin: restart API with `DEV_USER_EMAIL=ai@aegisimaging.ai` (or default) — all buttons return

7p. Defacing Service Backends

The defacing service supports multiple pluggable backends selected via the `DEFACE_TOOL` env var. DeepDefacer (3D U-Net) is the recommended production default for speed.

Local verification (without Docker)

- ☐ Install dependencies:

```
cd defacing
pip install -r requirements.txt
```

```
pip install deepdefacer
```

- ☐ Start with DeepDefacer:

```
DEFACE_T00L=deepdefacer uvicorn app.main:app --port 8081
```

- ☐ Verify health endpoint:

```
curl http://localhost:8081/healthz | python3 -m json.tool  
# → {"status":"ok","backend":"deepdefacer","available":true}
```

- ☐ Test explicit backend selection:
 - `DEFACE_T00L=nibabel` → health shows `nibabel-fallback`
 - `DEFACE_T00L=mri_deface` → falls back to `nibabel-fallback` (unless `mri_deface` installed)
 - `DEFACE_T00L=auto` → uses first available (deepdefacer if installed)

Docker verification

- ☐ Build with DeepDefacer:

```
docker compose build defacing  
# docker-compose.yml sets INCLUDE_DEEPDEFACER=true by default
```

- ☐ Start defacing service:

```
docker compose up defacing
```

- ☐ Verify via API health: `curl http://localhost:8080/healthz | python3 -m json.tool` → `services.defacing` shows healthy

7q2. Cloud AI Backends (PHI Detection + Classification)

Both the PHI detection and classification services support pluggable cloud AI backends. Cloud backends are optional — the services work with local-only backends (Tesseract, heuristic) by default. Cloud backends offer better accuracy for edge cases (burned-in text in poor image quality, missing DICOM tags).

GCP Cloud Vision setup

- ☐ Enable the Cloud Vision API in your GCP project:

```
gcloud services enable vision.googleapis.com
```

- ☐ Create a service account for AEGIS:

```
gcloud iam service-accounts create aegis-vision \  
  --display-name="AEGIS Cloud Vision"  
gcloud projects add-iam-policy-binding $PROJECT_ID \  
  --member="serviceAccount:aegis-vision@$PROJECT_ID.iam.gserviceaccount.com"
```

```
\
--role="roles/aiplatform.user"
```

- ☐ Download the key (local dev only — use workload identity in production):

```
mkdir -p secrets
gcloud iam service-accounts keys create secrets/gcp-key.json \
  --iam-account=aegis-vision@$PROJECT_ID.iam.gserviceaccount.com
```

- ☐ Set env var: `export GOOGLE_APPLICATION_CREDENTIALS=./secrets/gcp-key.json`
- ☐ Build Docker images with Cloud Vision:

```
INCLUDE_GOOGLE_VISION=true docker compose build phi-detection classification-
service
```

- ☐ Start and verify:

```
docker compose up -d phi-detection classification-service
curl -s http://localhost:8082/healthz | python3 -m json.tool # → backend:
google_vision
curl -s http://localhost:8085/healthz | python3 -m json.tool # → backend:
google_vision
```

AWS Textract / Rekognition setup

- ☐ Create an IAM user or role with these policies:
 - PHI detection: `AmazonTextractFullAccess`
 - Classification: `AmazonRekognitionReadOnlyAccess`
- ☐ Set AWS credentials:

```
export AWS_ACCESS_KEY_ID=<your-key>
export AWS_SECRET_ACCESS_KEY=<your-secret>
export AWS_DEFAULT_REGION=us-east-1
```

- ☐ Build Docker images with AWS backends:

```
INCLUDE_AWS_TEXTRACT=true INCLUDE_AWS_REKOGNITION=true docker compose build
phi-detection classification-service
```

- ☐ Override backend selection (if you want AWS instead of auto):

```
PHI_T00L=aws_textract CLASSIFY_T00L=aws_rekognition docker compose up -d phi-
detection classification-service
```

Verify cloud backend selection

- ☐ Check health endpoints after starting services:

```
curl -s http://localhost:8082/healthz | python3 -m json.tool # PHI detection
curl -s http://localhost:8085/healthz | python3 -m json.tool #
classification
```

- ☐ Backend field shows the active backend (`google_vision` , `aws_textract` , `aws_rekognition` , `tesseract` , or `heuristic`)
- ☐ Auto mode selects: PHI: `google_vision` > `aws_textract` > `tesseract`. Classification: `google_vision` > `aws_rekognition` > `heuristic`.
- ☐ Upload a study and trigger PHI scan / classification → results should come from the cloud backend
- ☐ Test fallback: unset cloud credentials → restart → falls back to local backend

7r. Study Export & DICOM Download

Admin DICOM Download

- ☐ Upload and approve a study
- ☐ In the admin dashboard, click **Download DICOM** on the approved study row
- ☐ Verify a zip file downloads containing all DICOM files (`.dcm`)
- ☐ Verify unapproved studies do NOT show the Download DICOM button

Export Portal (Share Recipient Download)

- ☐ Create an export share for an approved study (admin dashboard → Share panel)
- ☐ Copy the share token from the API response
- ☐ Open the export portal: `http://localhost:3004?token={token}`
- ☐ Verify study info is displayed: modality, body part, file count, description, expiry
- ☐ Click **Download All (ZIP)** → zip file downloads
- ☐ Test expired/revoked token → clean error message displayed
- ☐ Test invalid token → "Share not found" error

Export Forwarding (DICOMweb STOW-RS)

- ☐ Open admin dashboard → **Routing** tab
- ☐ Create a Destination (type: `dicomweb` , URL of a DICOMweb endpoint)
- ☐ Create a Routing Rule: action `require_export` + `route_to` (select the destination)
- ☐ Upload a study → verify `export_required=true` and `export_status=pending` in the studies table
- ☐ Approve the study → verify export auto-dispatches (status changes to `exporting` → `exported`)
- ☐ Check Audit Log tab → `export.triggered` and `export.complete` entries appear
- ☐ Test manual re-trigger: click **Export** button on a failed export → retries forwarding

Export Portal Dev Server

```
cd frontend/export-portal && npm install && npm run dev # runs on :3004, proxies
/api to :8080
```

7q. Automated Processing Pipeline

The pipeline auto-dispatches processing services after upload. Enabled by default (`PIPELINE_AUTO=true`).

- ☐ Configure routing rules that require multiple services (e.g. `require_classification + require_defacing + require_qc_check + require_bids_conversion`)
- ☐ Upload a study via the upload portal
- ☐ Check API logs for `pipeline: dispatching classification for ...` — classification runs first
- ☐ After classification completes, verify logs show parallel dispatch of defacing + PHI scan (if configured)
- ☐ After defacing completes, verify QC check and BIDS conversion are auto-dispatched
- ☐ Verify audit log shows `pipeline.dispatch` entries for each service
- ☐ Test manual override: click a manual trigger button in the dashboard — should still work
- ☐ Test disable: set `PIPELINE_AUTO=false` , upload again — no auto-dispatch, manual buttons required

7r2. Study Detail Page

The admin dashboard now includes a study detail view. Clicking a study UID in the studies table opens a dedicated panel.

- ☐ Open admin dashboard: `http://localhost:3001`
- ☐ Click any study UID link in the studies table → detail panel opens
- ☐ Verify header shows full study UID, status badge, source badge, description
- ☐ Verify meta row shows modality, body part, files, series, store, timestamps
- ☐ Verify pipeline visualization shows 7 stages with color-coded dots
- ☐ Test action buttons: Approve, Reject, Classify, Scan for PHI, etc.
- ☐ Click "View" → Weasis viewer opens inline
- ☐ For approved studies: verify share form appears, create a share link
- ☐ Check Audit Trail tab → shows all audit entries for this study
- ☐ Check Routing Log tab → shows routing rule evaluations
- ☐ Check Shares tab → shows created export shares
- ☐ Click "← Back to studies" → returns to the studies table
- ☐ API endpoints: `GET /api/studies/{id}` and `GET /api/studies/{id}/audit`

7s. Automated Tests (Go API)

Three-tier test suite: unit tests (no Docker), model integration tests (real PostgreSQL via testcontainers), and handler HTTP tests (httptest + real DB + temp storage).

Unit tests only (no Docker required)

- ☐ Run: `cd api && go test -short -v ./...`
- ☐ Verify ~55 tests pass: routing engine, auth JWT parsing, CORS, config, email templates, storage, slugify

Full test suite (requires Docker)

- ☐ Ensure Docker Desktop is running
- ☐ Run: `cd api && go test -v -count=1 ./...`
- ☐ Verify ~120 tests pass including model CRUD and handler HTTP tests
- ☐ Run with race detector: `cd api && go test -race -count=1 ./...`

Test helpers (`api/testutil/`)

- `TestDB(t)` — spins up PostgreSQL 15 via testcontainers, runs migrations, auto-cleanup
- `TestServer(t, db)` — creates `handler.Server` with temp local storage, pipeline disabled
- `SeedProject(t, db) / CreateTestStudy(t, db, projectID) / CreateTestAdminUser(t, db, email, role)` — test fixtures

Makefile targets

```
make test      # full suite (requires Docker)
make test-unit # unit tests only (no Docker)
make test-race # full suite with race detector
```

8. Go API

- ☐ `cd api && go run .` — verify health endpoint at <http://localhost:8080/healthz>
- ☐ Add required Go dependencies as needed (storage SDKs, auth providers)
- ☐ Implement signed URL generation endpoint
- ☐ Test upload flow: browser → signed URL → GCS staging bucket

8a. Local Services (Docker Compose)

`docker-compose.yml` starts the **full platform stack** — database, email, viewer, Go API, and all Python services:

- ☐ `docker compose up -d` — builds and starts all 11 services
- ☐ Verify API health: `curl http://localhost:8080/healthz | python3 -m json.tool`
 - Should show `"status":"ok"`, `"database":"healthy"`, `"storage":"healthy"`, and all configured sidecar services as `"healthy"`
- ☐ Verify Weasis loads at <http://localhost:3005>
- ☐ Verify Mailpit web UI at <http://localhost:8025>

Services started by `docker compose up` :

Service	Port	Notes
postgres	5432	Data in pgdata named volume (persists across restarts)
mailpit	1025 / 8025	SMTP capture + web UI
weasis	3005	Weasis DWV viewer
api	8080	Go API (runs migrations on startup)

defacing	(internal)	Defacing service
phi-detection	(internal)	Burned-in PHI detection
qc-service	(internal)	QC automation
bids-service	(internal)	NIfTI/BIDS conversion
classification-service	(internal)	Metadata classification
protocol-service	(internal)	MRI protocol compliance
dimse-receiver	11112 (DICOM), 8080 (internal health)	DIMSE C-STORE ingress adapter

Start individual services:

```
docker compose up -d postgres      # just database
docker compose up -d api           # API + postgres (auto-dependency)
docker compose build               # rebuild all images after code changes
docker compose down -v            # stop + destroy volumes (fresh start)
```

Note: The frontends (upload-portal on :3000, admin-dashboard on :3001) still run via `npm run dev` outside Docker, proxying `/api` to `localhost:8080`.

8aa. DIMSE PACS E2E Validation Harness

- ☐ Install DIMSE receiver dependencies:

```
cd dimse-receiver && pip install -r requirements.txt -r requirements-test.txt
```

- ☐ Run harness:

```
python3 scripts/dimse_pacs_e2e_harness.py
```

- ☐ Verify all four scenarios pass:
 - `success_c_store_ingest`
 - `transient_failure_to_retry_queue`
 - `process_controls_restore_ingestion`
 - `dead_letter_path_and_recovery`
- ☐ Admin dashboard verification:
 - Open Admin Dashboard → **DIMSE Ops** tab
 - Verify summary cards show pending/dead-letter counters
 - Trigger one control action (for example, `Process due`) and verify counters/actions refresh
- ☐ Alerting verification (optional but recommended):
 - Set `DIMSE_RETRY_ALERTS_ENABLED=true`

- Configure at least one threshold (`DIMSE_RETRY_ALERT_DEAD_LETTER_NONZERO=true` or age thresholds)
- Trigger threshold condition and verify `GET /ingest/retry/alerts` returns alert entries
- Verify Admin Dashboard **DIMSE Ops** tab shows **Recent Retry Alerts**
- ☐ Verify durable retry state (restart-safe):
 - Ensure `DIMSE_INGEST_DURABLE_STORE_ENABLED=true` (default in `docker-compose.yml`)
 - Confirm state file path is on shared volume (`DIMSE_INGEST_DURABLE_STORE_PATH` , default `/app/data/dimse-ingest-retry-state.json`)
 - Create at least one pending/dead-letter entry, restart `dimse-receiver` , and verify `/ingest/retry` counters persist
- ☐ On failure, review printed `dimse-receiver` log tail and rerun with `--keep-logs`
- ☐ Save run output as pilot evidence

Runbook:

- `docs/planning/dimse-pacs-e2e-validation-runbook.md`
- `docs/dicom-conformance.md` — DICOM conformance statement (SOP classes, transfer syntaxes, DICOMweb services, DIMSE services, de-identification profile)

8ab. DICOM File Retention & Cleanup Policy

File lifecycle

DICOM files are written to shared storage at:

```
dicom/raw/{studyInstanceUID}/{index}.dcm    ← original tag-de-identified files
dicom/clean/{studyInstanceUID}/{index}.dcm  ← defaced output (when defacing is
required)
```

The **DIMSE receiver** writes to `dicom/raw/` when it receives a C-STORE. The **upload portal** and **batch import CLI** also write to `dicom/raw/` . These files are the canonical storage copy — no automatic cleanup runs. Files accumulate until explicitly managed.

When is it safe to delete?

Studies in a terminal state are safe to archive or delete:

Status	Safe to delete raw files?	Notes
approved	Yes	Defaced copy in <code>dicom/clean/</code> if defacing was required
rejected	Yes	No further processing will occur
defaced	Raw only	Keep <code>dicom/clean/</code> until approved or rejected
received / defacing / clean	No	Still in active pipeline

Do **not** delete files for studies still in the processing pipeline — the sidecars read from shared storage and will fail with missing-file errors.

Production cleanup (GCS)

Set a GCS object lifecycle rule on the DICOM bucket so raw files transition automatically:

```
# Create lifecycle config (adjust age to your retention policy)
cat > /tmp/lifecycle.json << 'EOF'
{
  "lifecycle": {
    "rule": [
      {
        "action": { "type": "SetStorageClass", "storageClass": "COLDLINE" },
        "condition": { "age": 90, "matchesPrefix": ["dicom/raw/"] }
      },
      {
        "action": { "type": "SetStorageClass", "storageClass": "COLDLINE" },
        "condition": { "age": 365, "matchesPrefix": ["dicom/clean/"] }
      }
    ]
  }
}
EOF
gsutil lifecycle set /tmp/lifecycle.json gs://YOUR_GCS_BUCKET
```

For hard deletion instead of Coldline transition, change `"type": "Delete"` and remove `"storageClass"`.

Note: GCS lifecycle rules apply only to object age, not study status. Coordinate retention periods with your data governance policy and any IRB or DUA requirements.

Production cleanup (S3)

```
aws s3api put-bucket-lifecycle-configuration \
  --bucket YOUR_S3_BUCKET \
  --lifecycle-configuration '{
    "Rules": [
      {
        "ID": "dicom-raw-coldline",
        "Filter": { "Prefix": "dicom/raw/" },
        "Status": "Enabled",
        "Transitions": [{ "Days": 90, "StorageClass": "GLACIER_IR" }]
      }
    ]
  }'
```

Local dev cleanup

```
# Remove all DICOM files for one study (safe once approved/rejected)
rm -rf ./data/dicom/raw/{studyUID}
rm -rf ./data/dicom/clean/{studyUID}

# Full reset (destroys all data including PostgreSQL)
docker compose down -v

# BIDS output cleanup
rm -rf ./data/bids/{studyUID}
```

DIMSE retry state file

The durable retry state file (`dimse-ingest-retry-state.json` , default path `/app/data/dimse-ingest-retry-state.json`) is small and safe to leave in place. It is overwritten on each retry worker pass. Delete it only to reset the queue to empty (pending/dead-letter entries are lost):

```
rm /app/data/dimse-ingest-retry-state.json
# Then restart dimse-receiver - it starts with empty queues
```

8b. Email (Local Dev with Mailpit)

Email is disabled by default — all calls are silent no-ops when `SMTP_HOST` is unset. Mailpit is included in `docker-compose.yml` (step 8a). To run standalone:

Email is disabled by default — all calls are silent no-ops when `SMTP_HOST` is unset.

- ☐ Run Mailpit (or use `docker compose up -d mailpit`):

```
docker run -p 1025:1025 -p 8025:8025 axllent/mailpit
```

- ☐ Start the API with SMTP env vars:

```
cd api && SMTP_HOST=localhost SMTP_PORT=1025 go run .
```

- ☐ Open Mailpit web UI at <http://localhost:8025>
- ☐ Trigger each notification to verify:
 - Upload a study via the portal — enter your email in the "Your email (optional)" field on the preview screen
 - Approve the study in the admin dashboard → uploader gets approval email
 - Reject a study → uploader gets rejection email
 - Create a share for an approved study → recipient gets share email with download link
- ☐ Verify no crash when `SMTP_HOST` is unset (email silently skipped, no error returned)
- ☐ Verify upload with no email entered still completes successfully (no notification sent)

8c. Landing Page & Contact Form (Vercel + Brevo)

Site: aegisimaging.ai | **Source:** `frontend/landing/` | **Framework:** Vite + React

The landing page is deployed to Vercel as a static site with a serverless function for the contact form. Vercel auto-deploys on every push to the connected branch — no manual deploys needed after initial setup. Every PR also gets a preview URL.

Step 1: Set up Vercel project

- ☐ Create a Vercel account for AEGIS Imaging LLC (separate from Encore) at vercel.com (free Hobby plan)
- ☐ Click **Add New → Project**
- ☐ Connect your GitHub account and select the **AEGIS** repo
- ☐ **Configure Project:**

Setting	Value
Root Directory	frontend/landing
Framework Preset	Vite (should auto-detect)
Build Command	npm run build (default)
Output Directory	dist (default)

- ☐ Click **Deploy** — wait for the first build to succeed
- ☐ Note the preview URL Vercel gives you (e.g. `aegis-abc123.vercel.app`)

Step 2: Configure contact form email (Brevo SMTP)

The contact form uses a Vercel serverless function (`api/contact.ts`) that sends email via Brevo SMTP. Without these env vars, submissions are logged to the console but not emailed — you can configure this later.

- ☐ Create a Brevo account for AEGIS Imaging LLC at brevo.com (free tier: 300 emails/day)
- ☐ Go to **Settings → SMTP & API → SMTP** and note your credentials
- ☐ In your Vercel project: **Settings → Environment Variables**, add:

Key	Value	Environments
BREVO_SMTP_HOST	smtp-relay.brevo.com	Production, Preview
BREVO_SMTP_PORT	587	Production, Preview
BREVO_SMTP_USER	(your Brevo SMTP login)	Production, Preview
BREVO_SMTP_PASS	(your Brevo SMTP password)	Production, Preview
BREVO_SMTP_FROM	AEGIS <noreply@aegisimaging.ai>	Production, Preview

- ☐ Redeploy (Settings → Deployments → click : on latest → **Redeploy**)
- ☐ Test: submit the contact form → check inbox at `contact@aegisimaging.ai`

Step 3: Add custom domain in Vercel

- ☐ In your Vercel project: **Settings → Domains**
- ☐ Type `aegisimaging.ai` and click **Add**

- ☐ Vercel will show the DNS records you need — keep this page open

Step 4: Configure DNS at GoDaddy

- ☐ Log in to godaddy.com
- ☐ Go to **My Products** → **aegisimaging.ai** → **DNS** → **Manage DNS**
- ☐ Delete any GoDaddy parking/forwarding records if present
- ☐ Add or edit these records:

Type	Name	Value	TTL
A	@	76.76.21.21	1 Hour
CNAME	www	cname.vercel-dns.com	1 Hour

- ☐ Save changes
- ☐ Go back to the Vercel Domains page — wait for the green checkmark (5–30 min, sometimes up to 48 hours)
- ☐ Vercel provisions a free SSL certificate automatically

Step 5: Set up aegisimaging.org redirect (optional)

- ☐ **Option A — Via Vercel:** Add `aegisimaging.org` as a domain in the same Vercel project. Vercel will redirect it to the primary domain.
- ☐ **Option B — Via GoDaddy:** On the `.org` domain, set up a domain forward: GoDaddy → My Products → aegisimaging.org → Manage → Forwarding → Forward to `https://aegisimaging.ai`

Step 6: Verify everything works

- ☐ Visit `https://aegisimaging.ai` — page loads with SSL
- ☐ Visit `https://www.aegisimaging.ai` — redirects to apex
- ☐ Scroll through all sections — animations trigger
- ☐ Test mobile layout (resize browser or use phone)
- ☐ Submit the contact form — check inbox at `contact@aegisimaging.ai`
- ☐ Visit `https://aegisimaging.org` — redirects to `.ai` (if configured)

How the contact form works

The contact form (`POST /api/contact`) has two independent delivery paths:

Path	Backend	When
Vercel serverless function	<code>frontend/landing/api/contact.ts</code> — nodemailer + Brevo SMTP	Landing page on Vercel (standalone)
Go API endpoint	<code>api/handler/contact.go</code> — existing SMTP config	Landing page proxied to Go backend

Both accept `{ name, email, organization, role, message }` and return `{ sent: true }`. The Vercel function is self-contained — it doesn't depend on the Go API. This means the landing page and contact form work even before the backend is deployed.

9. GitHub Repository

Organization setup

Use GitHub Organizations to separate codebases by company.

- ☐ Create GitHub Organization: `aegis-imaging`
- ☒ Repo at `aegis-imaging/aegis` — transfer complete
- ☒ Local remote: `git@github.com:aegis-imaging/aegis.git`

Repository setup

- ☐ Verify remote is set: `git remote -v`
- ☐ Push monorepo scaffold to `develop` branch
- ☐ Set up branch protection on `main` and `develop` (require PR reviews, prevent deletion)
 - **Requires GitHub Pro** (\$4/month) for private repos, or make the repo public
 - Go to Settings → Branches → Add branch protection rule
 - Branch name patterns: `main` and `develop`
 - Enable: "Require a pull request before merging", "Do not allow deletions"
- ☐ CI is configured: `.github/workflows/ci.yml` runs automatically on PRs to `develop` and `main`
 - Go build + vet, Python syntax check (7 services), TypeScript type check (5 apps), Docker build (8 images)
- ☐ Verify CI passes: open a test PR and check the Actions tab
- ☐ Run `make lint` locally to validate before pushing

10. Future — Before Proposing to Work

- ☐ Have a working end-to-end demo: upload → anonymize → view in Weasis
- ☐ Prepare a 5-minute screen recording of the demo flow
- ☐ Draft a one-page proposal covering: problem, solution, differentiation, cost estimate
- ☐ Identify potential pilot users / departments at your institution
- ☐ Research your company's internal startup / innovation program requirements

Generated 2026-02-18. Updated 2026-02-23. See `AEGIS_Architecture.md` for the full system design. DIMSE receiver Compute Engine VM, Cloud Build CI/CD triggers, IAM hardening, DICOM conformance statement, and alert runbooks reflected through 2026-02-23.